Capstone Project Proposal Report
(Individual Report)

Instructions:
This form is to be completed by each student doing Project registration to fulfill their senior design or capstone requirement. It must be completed and submitted to your Guide. Each student must complete this form individually.
This report is to be completed during the starting of the semester, while the project description report will be completed during end of the semester.

Guide Approval (initials/date):		

CAP4001– Capstone Project Proposal Report

Student Name	**Desam Pranay Kumar Reddy**
Student Register Number	**23BCE8722**
Programme	**CSE AI/ML**
Semester/Year	**7th Semester**
Guide(s)	**Balusa Bhanu Chander**
Project Title	**An Explainable AI-Based Decision Support System for Dynamic Water Quality Index Prediction and Recommendation System**

Team Composition: Provide the information below for each member of the **project team**. Include **all** project team members, not just those in your discipline or those enrolled for Capstone project. Please also include yourself!

Reg. No	Name	Major	Specialization
23BCE9594	Chintakayala Karthikeya	CSE	AI/ML
23BCE7692	Ameer Roshan Shaik	CSE	AI/ML
23BCE9533	Shaik Abdul Samad	CSE	AI/ML
23BCE8722	Desam Pranay Kumar Reddy	CSE	AI/ML

Project and Task Description:

(a) Project Summary

Water quality assessment helps in providing the safe water, protecting the environment, and managing the water resources in an effective manner. Now the process of determining the Water Quality Index (WQI) is done manually by considering the number of water quality parameters. It is therefore not effective for large-scale implementations.

The current study aims to develop an Explainable Artificial Intelligence (XAI) based Decision Support System for predicting Water Quality Index (WQI) using an appropriate Machine Learning Algorithm. The project will adopt 2000 water quality instances retrieved from water monitoring stations as the target dataset. Water quality parameters that will be considered include Temperature, Dissolved Oxygen, pH, Conductivity, Biological Oxygen Demand (BOD), Nitrate + Nitrite , Fecal Coliform, and Total Coliform.

The project will calculate Water Quality Index using a published standard methodology applicable in the target dataset. The project will develop a number of machine learning models including Random Forest, XGBoost, and LightGBM. The best performing model will be interpreted using SHAP values to understand the relationship between the input parameters and the predicted WQI. The project will make recommendations on water quality based on the predicted WQI.

(b) Individual Role and Tasks

As an individual member of project team, I will primarily contribute to **Explainable Artificial Intelligence and model interpretation stages** of the project. My responsibilities include:

- Conduct literature review on Explainable Artificial Intelligence techniques for the machine-learning models.
- Studying the performance of the developed prediction models and identify suitable models for detailed interpretation.
- Implementing SHAP (SHapley Additive exPlanations) for the explaining model predictions.
- Analyze the global feature importance to identify water-quality parameters that most strongly influence predicted WQI.
- Generate SHAP summary plots and the feature-contribution visualizations.
- Perform the local explanation of the individual predictions.
- Compare important feature patterns across the selected models and datasets.
- Analyze relationship between water-quality parameters and their positive or negative contribution to predicted WQI.
- Support interpretation of the model results from an environmental and water-quality perspective.
- Prepare visualizations and explanation results for research paper and final presentation.

Individual Deliverables

Literature Review Contribution
Explainable AI Methodology
SHAP Implementation Module
Global Feature Importance Analysis
SHAP Summary Plots
Feature Contribution Visualizations
Individual Prediction Explanations
Model Interpretation Report
Explainability Results and Discussion
Research Paper Contribution
Final Project Documentation
Presentation Slides

(c) Specific Approach

The given project will adopt the standard approach used in machine learning. First of all, the water quality data set will be collected and preprocessed. It includes removing missing values and other irrelevant information to prepare the data for training.

Then, with the use of the published method for calculating the Water Quality Index (WQI), that is applicable to the given data set, the WQI will be determined. Following that, the Exploratory Data Analysis will be performed to examine the relationship between various parameters and detect patterns in the data set.

After that, feature engineering will be applied, and various algorithms, among which Random Forest, XGBoost, and Ligther GBM, will be trained on the given data set. The performance of all the models will be compared using evaluation metrics such as Mean Absolute Error, Root Mean Square Error, and R2 Score. Then, the performance of the best model will be explained using the SHAP (SHapley Additive exPlanations). In the end, a Decision Support System will be designed that will indicate the predicted WQI and give a recommended statement on the quality of water based on the prediction.

(d) Design Phases

Phase Activities

Phase 1 Literature Review and Problem Identification
Phase 2 Dataset Collection and Data Preprocessing
Phase 3 Water Quality Index Calculation
Phase 4 Exploratory Data Analysis and Feature Engineering
Phase 5 Machine Learning Model Development
Phase 6 Model Training, Hyperparameter Tuning and Evaluation
Phase 7 Explainable AI Implementation using SHAP
Phase 8 Decision Support System Development
Phase 9 Testing, Documentation, Research Paper Preparation and Final Presentation

Outcome Matrix: Describe your plan to demonstrate each of the outcomes below.

Outcomes:	**Plan for demonstrating outcome:**
a) an ability to apply knowledge of mathematics, science, and engineering	Apply machine-learning interpretation, statistics, feature analysis, and environmental water-quality concepts to understand how individual parameters influence predicted Water Quality Index values
c) an ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	Design an explainability workflow that provides understandable model interpretations while considering the complexity of predictive models, reliability of explanations, computational resources, and the need for meaningful environmental interpretation
d) an ability to function on multidisciplinary teams	Collaborate with team members by interpreting the predictions generated by the modelling component and providing feature-contribution insights for the decision-support system.
e) an ability to identify, formulate, and solve engineering problems	Address the problem of limited interpretability in complex machine-learning models by applying Explainable Artificial Intelligence techniques to explain model predictions and feature contributions.
g) an ability to communicate effectively	Communicate model interpretation results using the SHAP summary plots, feature-importance visualizations, individual prediction explanations, reports, research paper content, and the presentations.
k) an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice	Use Python, SHAP, Pandas, NumPy, Scikit-learn, Matplotlib, and the machine-learning frameworks to generate and analyze model explanations and visualizations.

Realistic Constraints:

There are various realistic constraints taken into consideration in the project including lack or imperfection of water quality readings, variance in water quality at various monitoring stations, computing constraints during model training, and availability of certain water quality parameters only for calculation of the Water Quality Index. Environmental sustainability, proper use of open-source data and the generation of reliable recommendations is also taken into account.

Engineering Standards:

The project will comply with the established engineering and research standards throughout its implementation. Standard machine learning techniques like data pre-processing, feature engineering, validation of models, hyperparameter tuning and evaluation of the models with MAE, RMSE and R^2 Score will be used. Explainable Artificial Intelligence (XAI) will be utilized via SHAP. The Decision Support System will provide recommendations on the basis of WQI values according to certain water quality categories. Documentation and research reporting will be done according to IEEE standards.